



**Corporate Average Fuel Economy (CAFE)
Database (DB)**
Systems Development and Implementation

STATEMENT OF OBJECTIVES (SOO) DRAFT

**U.S. Department of Transportation
National Highway Traffic Safety Administration**

SECTION C. CAFE DB Statement of Objectives (SOO) (DRAFT)

1. Introduction

This Statement of Objectives (SOO) defines the requirements for the continued Development, Modernization, Enhancement (DME), and Operations and Maintenance (O&M) of the Corporate Average Fuel Economy (CAFE) Light Duty (LDV) and Medium Duty/Heavy Duty (MD/HD) Management Suite, hereafter referred to as the **CAFE DB**.

2. Background

The Department of Transportation (DOT), National Highway Traffic Safety Administration (NHTSA) fuel efficiency programs are governed by 49 USC 329. The fuel efficiency programs require manufacturers of Light, Medium, and Heavy-Duty vehicles (LDV and MDV/HDV), produced for sale in the US, to meet fuel efficiency standards, expressed in miles per gallon (mpg) for LDV vehicles and fuel consumption standards (gal/ton-mi) for MDV/HDV Class 2b through Class 8 vehicles.

The purpose of these programs is to reduce the nation's energy consumption through increasing the fuel efficiency of motor vehicles. Fuel efficiency standards improve our nation's energy security and save consumers' money at the pump. NHTSA establishes LDV standards (domestic and import passenger cars, and light trucks) and MDV/HDV standards at "the maximum feasible fuel efficiency levels." Manufacturers must ensure their annual fleet averages meet established standards using various technologies, and the CAFE DB tracks these credits and compliance balances.

System Description:

The system acts as a centralized repository for the secure receipt of data from Original Equipment Manufacturers (OEMs). Data is currently ingested through two primary channels: **Standardized Excel templates** submitted via email and the **NHTSA/EPA web service**. This data includes mid- and final-model year reports, fuel economy data, vehicle footprints, and production volumes.

CAFE currently serves as an easy and efficient means of sharing regulated fuel efficiency data collaboratively with the Environmental Protection Agency (EPA) Central Data Exchange (CDX). The CMS provides a simplified means of providing up-to-date reoccurring public reporting on manufacturers' efforts in complying with fuel efficiency standards.

The CAFE DB is categorized as a **FIPS 199 Moderate** system. It is the primary technological infrastructure supporting the rulemaking, enforcement, and regulatory reporting mandates for NHTSA's Office of Vehicle Safety Compliance (OVSC) and Office of Rulemaking (NRM).

Primary Functional Areas

The CAFE DB is organized into four core functional modules:

- **Light Duty Vehicles (LD) Management:** Administers compliance and data tracking for passenger cars and light trucks.
- **Medium Duty/Heavy Duty Vehicles (MD/HD) Management:** Manages fuel consumption standards for Class 2b through Class 8 vehicles.
- **Business Intelligence (BI) Reporting:** Provides advanced data visualization and analytical tools for internal NHTSA stakeholders to track compliance trends.
- **Public Information Center (PIC):** Powers the public-facing portal where fuel efficiency data and manufacturer compliance efforts are reported for transparency. Available at https://one.nhtsa.gov/cape_pic/cape_pic_home.htm.

Administrative & Credit Functions

Beyond data ingestion, the system automates the following regulatory workflows:

- **Credit Administration:** Manages and records credit trades and transfers between manufacturers. CAFE is indexed and searched by credit holder/OEM to access fleet/credit information, and not by an individual credit holder name.
- **Account Balancing:** Automatically applies or subtracts credits from OEM accounts to determine program compliance.
- **Enforcement Support:** Identifies negative credit balances and facilitates the issuance of fine letters.

3. Purpose

The purpose of this contract is to provide a modernized, cloud-native (AWS) application and a secure web-based interface that streamlines the lifecycle of fuel economy data management. This effort eliminates manual, error-prone processes by providing high-integrity data analytical tools for NHTSA and public reporting via the Public Information Center (PIC).

4. Acronyms

Acronym	Definition
AD	Active Directory
AI	Artificial Intelligence

Acronym	Definition
ASDLC	Agile Software Development Lifecycle
ATO	Authorization to Operate
AWS	Amazon Web Services
BI	Business Intelligence
BPA	Blanket Purchase Agreement
CAFE	Corporate Average Fuel Economy
CBI	Confidential Business Information
CDX	Central Data Exchange (EPA)
CI/CD	Continuous Integration / Continuous Delivery
CO	Contracting Officer
COR	Contracting Officer Representative
DME	Development, Modernization, and Enhancement
DOT	Department of Transportation
EOY	End of Year
EPA	Environmental Protection Agency
FIPS	Federal Information Processing Standards
HDV	Heavy-Duty Vehicle
LDV	Light-Duty Vehicle
LVM	Low Volume Manufacturer
MDV	Medium-Duty Vehicle

Acronym	Definition
MFR	Monthly Financial Report
ML	Machine Learning
NHTSA	National Highway Traffic Safety Administration
NRM	Office of Rulemaking
O&M	Operations and Maintenance
OEM	Original Equipment Manufacturer
OVSC	Office of Vehicle Safety Compliance
PIA	Privacy Impact Assessment
PIC	Public Information Center
PIV	Personal Identity Verification
PMP	Program Management Plan
QASP	Quality Assurance Surveillance Plan
SOO	Statement of Objectives
SOP	Standard Operating Procedure
SSP	System Security Plan
SWES	Software Engineering Services
UAT	User Acceptance Testing

5. Scope

The government has identified a need for technically innovative support services to provide a holistic solution to build a modernized fuel economy data management suite that streamlines the lifecycle of regulatory data across the enterprise. Some key

characteristics desired for the modernized platform and implementation services are as follows:

Req ID	Requirement	Expanded Operational Explanation
1	Web-based Reporting Interface	Design of a secure external portal for direct OEM data submission to streamline the data collection and reporting processes and improve data validation and verification features.
2	Class 2b-3 Final Reporting	Development of Heavy-Duty Pickup Trucks and Vans (HDPUV) loading functionality to incorporate model type data.
3	CBI Treatment Requests	Integration of CAFE DB with the Confidential Business Information (CBI) application to support Legal Archives and FOIA.
4	Industry Highlights Documents	Automation of Highlights Reports as a user-guided report generation process used to provide high-level visibility into fuel economy performance, trends, and compliance for the agency and external stakeholders.
5	90-day EOY Reporting	Development of an end-of-year production data reporting module for Medium Duty/Heavy Duty (MD/HD) and HDPUV fleets.
6	Credit Transaction Templates	Development of a module to auto-generate credit transaction data from the standardize template and apply to CAFE data fields
7	Credit Plans	Migration of manufacturer credit planning documents from network drives into a traceable database format.
8	Credit Value Reporting	Creation of a centralized DB location for reporting and storing credit valuations.
9	Off-cycle Credit Requests	Migration of off-cycle (OC) technology credits to the DB for compliance tracking.

Req ID	Requirement	Expanded Operational Explanation
10	Corporate Relationship Status	Establishing a single location in the CAFE DB for checking on the status of corporate relationships.
11	Low Volume Status (LVM) Status	Implementation of a centralized module to verify if a manufacturer is exempt under low-volume status.
12	SPARKS Integration	Integration with the Safety Program & Record Keeping System (SPARKS) DB to enable cross-referencing of vehicle fuel efficiency with safety enforcement records and defect investigations.
13	PIC Reports Automation	Automated generation of standardized reports for public consumption.
14	Historical Data Migration	Loading of historical model year data into the DB.
15	Additional Priorities	As identified

Out of Scope

- Ground-up development of solutions outside existing DOT/NHTSA frameworks.
- Customizations requiring non-COTS development without prior authorization.
- Usage of non-approved information systems, tools, or proprietary data sources.
- Procurement of hardware or third-party data without written approval from the CO.

6. Performance Objectives and Deliverables

This Statement of Objectives (SOO) is intended to obtain responses in the form of a Performance Work Statement (PWS) providing each Offeror's proposed approach for the design, development, and implementation of the CAFE DB platform with the following anticipated outcomes:

1. **Modernize Data Ingestion:** Transition from manual Excel and email-based processes to a secure, web-based external ingestion portal for OEM data submission.

2. **Automate Regulatory Workflows:** Eliminate manual interventions for **90-day EOY Reporting, Credit Plans, and Credit Value Reporting**.
3. **Ensure Data Integrity:** Strictly segregate and protect **Confidential Business Information (CBI)** through a secure treatment request workflow and partitioned storage.
4. **Enhance Transparency:** Expand Business Intelligence (BI) reporting and automated data extractions for the **Public Information Center (PIC)** portal.
5. **Intelligence Integration:** Leverage **AI/ML** to identify data anomalies and improve the relevance of regulatory recommendations.
6. **Maintain High Availability:** Ensure **99.9% uptime** through robust Operations & Maintenance (O&M), including routine patching and incident mitigation within hours.
7. **Agile Engagement:** Support Agile methods by providing continuous integration/continuous delivery (CI/CD) capabilities, sprints, and retrospectives.
8. **Continuous Security Oversight:** Perform **Annual Security Assessments** and coordinate with the NHTSA Security Team to obtain and maintain a full **Authorization to Operate (ATO)** as required. *See Appendix B*

The Government will utilize a **Quality Assurance Surveillance Plan (QASP)** *see Appendix A* to monitor the Contractor's performance against the objectives and deliverables identified below.

The following table provides the expected deliverables and the details of the performance objectives and expectations for the **CAFE DB** platform.

Deliverable ID	Performance Objective / Deliverable Description
1	Provide the experienced technically-skilled staff to work independently, do the necessary research and analysis, follow up and follow through with the tasks, meetings, technical reviews with NHTSA and DOT staff (including DOT/OST contractor staff) to carry out the necessary duties and requirements to completely set up the platform.
2	Set up and follow through with the Agile methodology including but not limited to, daily standups, scrum meetings, and retrospectives, throughout the sprint lifecycles.
3	Within four (4) months after award, provide a Product Roadmap.

Deliverable ID	Performance Objective / Deliverable Description
4	Document and update milestones, tasks, and activities to provide a product status using the Product Status Report template and time schedule (weekly, bi-weekly).
5	Take meeting minutes for each working group, planning, and design meeting along with comments and action items. Provide minutes to attendees within three (3) business days of meeting.
6	Track and deliver Monthly Financial Status Reports (MFSR)— to track the "burn rate" and ensure the project remains within budget.
7	Use Confluence and Jira including the templates that NHTSA has created for standardization.
8	Identify risks, blockers, and obstacles causing schedule slippage to be reported in the Risk Register.
9	Develop and implement the CI/CD pipeline.
10	Present recommended solution(s) for the CAFE platform after gathering and assessing systems and requirements, business and systems impacts, priority weightings.
11	Successfully implement LDV & MD/HD compliance tracking, including Class 2b-3 reporting and model type data integration.
12	Implement a secure CBI Treatment Request workflow and partitioned storage to ensure sensitive data protection.
13	Provide experienced, technically skilled staff to conduct technical reviews and research with NHTSA and DOT staff.
14	Design and implement the External Ingestion Portal to modernize the receipt of regulatory data from OEMs.
15	Automate the tracking of Credit Plans, Credit Transaction Templates, and Credit Value Reporting.

Deliverable ID	Performance Objective / Deliverable Description
16	Meet cybersecurity and privacy requirements to obtain and maintain the Authorization to Operate (ATO) (including system security plans, privacy assessments and documentation, and other required cybersecurity documents and artifacts).
17	Provide users with a modernized and intuitive user interface (UI) where required to meet varying needs and levels of computer skills
18	Perform impact analysis and incur minimal disruption to the business, with consideration and guidance given by NHTSA's Enterprise / Data Architecture (NHTSA Architecture Review Board and NHTSA Enterprise Data Strategy).
19	Ensure user content collaboration and information sharing with multiple users and user groups accessing the system without overwriting each other's work.
20	Integrate environment administrative functions and capabilities with multiple users with concurrent access.
21	Implement role-based access controls (RBAC), user groups, (administrative functions and capabilities), and privileges to restrict system access based on job responsibilities and the functions necessary for users to effectively perform their research and reporting, and to support auditable tracking for the provisioned environments.
22	Provide and test performance with users and systems to ensure quality and meet performance standards and expectations.
23	Provide a solution that is scalable, resilient, and flexible.

7. Contract Performance

7.1 Place of Performance

The contract activity and tasks will be conducted off-site in accordance with the agreed-upon terms in the awarded BPA Call. Some meetings, tasks, and activities may need to occur at the

government facility located at 1200 New Jersey Avenue, Washington, DC only during the days when government operations are being conducted and with prior notice.

8. Government Information

- Comply with Digital Accountability and Transparency Act (DATA) of 2014 requirements.
- Comply with Federal Information Security Modernization Act (FISMA) of 2014.
- Comply with Federal Information Processing Standards (FIPS).
- Comply with the Guide to Protecting the Confidentiality of Personally Identifiable Information (PII).
- Comply with Section 508 accessibility guidelines.
- Meet security requirements to obtain and maintain the Authorization to Operate (ATO).
- Ensure compliance with the Artificial Intelligence (AI) Assessment documentation support under Executive Order 13960, "Promoting the Use of Trustworthy AI," and the recent Executive Order 14110, "Safe, Secure, and Trustworthy Development and Use of AI," and related OMB guidance M-24-10. See Appendix B for additional cybersecurity and privacy regulatory and policy requirements.

9. Government Property/Equipment

Government-furnished items may include, but are not limited to:

- Laptops with PIV-card compatibility
- Personal Identity Verification (PIV) cards
- AWS and Azure infrastructure
- DOT and NHTSA managed services such as Tableau, cloud services
- JIRA and Confluence applications for epics, stories, and meeting notes tracking.

10. Meetings

For all meetings, the contractor shall be responsible for providing meeting materials and administrative/facilitation support. Meetings will be conducted at either the NHTSA Headquarters – DC location at 1200 New Jersey Avenue, SE, Washington, DC, or through an alternative method of communication (such as a teleconference), as approved by the CO.

10.1 Post-Award Kickoff Meeting

Within 15- to 30 business days after contract award, the government will conduct a post-award kickoff meeting via teleconference (specific date and time to be mutually agreed upon).

10.2 Intermittent Product Status Reviews

At the sole discretion of the government, intermittent product status reviews may be conducted on an informal basis (by telephone or teleconference) with 5 days advanced notice as approved by the CO, COR, and/or PM.

10.3 Monthly Product Status Reviews

The contractor is responsible for reporting the previous month's activities (including any risks, issues, or concerns, and actual or recommended actions for their mitigation), and projected activities for the following month.

10.4 Program Management Review

The government will conduct bi-weekly program management meetings to review the progress of the program, identify any risks, issues, or concerns, and provide feedback on the contractor's progress and performance. Accomplishments for the reporting period (usually 2-3 weeks) and planned work are to be included in the product template for reporting. The contractor shall provide written data and verbal presentations as to the financial status of the CAFE program and related projects (projected vs. incurred costs), any identified any risks, issues, or concerns (and their mitigation or their plans for their mitigation).

11. Transition Planning

The Contractor shall deliver and execute a transition plan at a minimum of 45 days prior to the contract's end. The plan shall document the process for the transition of system support, software development tasks, activities, and all functions associated with the respective project, including the status of all deliverables and electronic delivery of all project code and artifacts. The Contractor shall transition documentation as listed in the delivery schedule and templates will be provided by the Federal Government, as required.

12. Notice to Contractor

All information and data related to this project that the contractor gathers or obtains shall be both protected from unauthorized release and considered the property of the government. The CO will be the sole authorized official to release verbally or in writing any data, draft deliverables, final deliverables, or any other written or printed materials pertaining to this contract. Press releases, marketing material, or any other printed or electronic documentation related to this project must not be publicized without the written approval of the CO.

(End of Section C)

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